

Equilibrium Router

Coupling frozen models through a shared anchor:
what the object is, and two pre-registered negatives

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github.com/gnostrich/Equilibrium-Router

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Abstract

Suppose you have several trained models that already work, that you cannot retrain, and that see the world differently — an image model, a text model, an audio model, a time-series model. The usual way to combine them is to average their outputs. Averaging is cheap and hard to beat, but it is also deaf: it returns the same answer whether the models agreed or fought.

We describe an alternative. Rather than averaging, we let the models pull on a shared state until that state stops moving, and we read the settling itself. The claim under test is that the way the system settles — smoothly, or with a persistent tremble — carries information that a single averaged number cannot.

We report what we built and what it did, including the parts that did not work. The object is genuinely a feedback system and not averaging in disguise: on deliberately conflicting inputs its coupled dynamics become unstable, which averaging is mathematically incapable of. It distinguishes agreement from conflict where averaging is blind. But on four real, independently trained models it settles quietly and ties the average exactly, and the payoff we were after — recovering competing interpretations from the tremble — performs at chance. Separately, a pre-registered measurement on an earlier anchor-based version of the system shows that its central diagnostic, while robust and reproducible, does not have the mathematical character we had ascribed to it: it is a metric quantity, not a topological one.

Both negatives were predicted in writing before the runs and are reported as primary results. We also record the one thing in the program that does work and does not need any of this machinery.

1 What this paper is

This paper documents a research object and reports honestly on what it does. It is not a results paper claiming a new state of the art; two of its four main findings are negative, and we think those are the most useful part.

The setting is model composition without retraining. Pre-trained models are cheap to obtain and expensive to modify. A practitioner with several frozen models — possibly across different modalities — wants their combined judgement. Averaging (or a learned linear readout on top of concatenated outputs) is the default and is a strong baseline.

Our interest was in a specific limitation of averaging. Averaging is a *summary*: it compresses several answers into one and discards the process by which they were reconciled. If three models arrive at the same answer for the same reason, and three others arrive at it after pulling hard in opposite directions, the average is identical. That discarded information — how much the models fought — is what we set out to recover.

We report three things:

- (i) **The object** (§3): models as forces tensioning a shared state, with the settling trace as the readout. Stated with intuition alongside the notation.
- (ii) **The design choices** (§4): the small number of decisions that determine such a system, which of them we settled and why, and which turned out to be determined by facts we only discovered by measuring.
- (iii) **What it does** (§5–§6): the measured behaviour, including two pre-registered negatives, and the current status of the program.

Every substantive claim carries a tag: [measured] for something we ran, [argued] for a structural argument without a run behind it, [open] for something we have not settled, and [retracted] for a claim we previously made and now withdraw.

2 Closest existing work

Our object sits at the intersection of three established lines, and we want to be precise about what is ours and what is not.

Composing frozen models. The observation that independently trained models learn compatible representations, and that a light map suffices to translate between them, is well established — the Platonic Representation Hypothesis [4] and relative representations [6] are the canonical statements, with model stitching [5, 2] as the older experimental version. Two recent systems are the closest in construction to ours: *Vision Wormhole* [9] and *Dead Weights, Live Signals* [10], which wires heterogeneous frozen language models as nodes of a feedforward graph communicating through a shared latent space via learned linear projections. Both are frame-dependent (they require or learn a correspondence between models’ coordinates) and task-supervised. Ours differs in being an equilibrium rather than a feedforward pass, and in communicating through transport geometry rather than learned projections — but we note that [10] reports as its own limitation that its projections fail to specialise, which is adjacent to the gap we were aiming at.

Equilibrium models. Defining a network’s output as the fixed point of a coupled update, rather than as the result of a fixed number of layers, is the deep equilibrium model [1]. We use the same idea, but the fixed point is over a shared state that several *frozen external* models write into, not over the weights of one trained network.

Transport geometry. The anchor-based version of our system (§6.1) represents each model by the internal geometry of its output cloud and couples models to a shared set of anchor points using entropic Gromov–Wasserstein transport [7, 8]. Gromov–Wasserstein is the standard tool for comparing point clouds that live in different spaces with no correspondence between their coordinates, which is exactly the cross-modal situation. Free-support barycenters [3] are the standard way to place the shared anchors.

What is ours. None of the ingredients. The combination — an equilibrium over a shared transport anchor with a diagnostic read *outside* the objective being minimised — is the thing we built, and this paper’s contribution is an honest account of how it behaves rather than a claim that the combination is valuable. On the evidence below, the strongest defensible statement is that the mechanism is real and its payoff does not materialise on real inputs.

3 The object

3.1 Intuition

Picture a taut sheet. Each model grabs the sheet in a different place and pulls in the direction it thinks is right. Let go, and the sheet settles. Three things can happen. It can come to rest at a point — the models agreed, or their disagreements cancelled. It can keep trembling — the models are pulling against each other and the tension never resolves. Or it can fly apart — the pulls reinforce each other and the whole thing goes unstable.

Averaging is the special case where you skip the sheet entirely and just take the mean of where everyone is pulling. It always gives you a point, and it can never fly apart. That last fact is the key one, and we use it as our test of whether we have actually built something different (§5.1).

3.2 Notation

Let there be K frozen models. Model i is represented by an operator R_i acting on a shared state $s \in \mathbb{R}^D$: given the current state, $R_i s$ is where model i would like the state to be. We call R_i model i 's *frame*. The state settles under

$$s \leftarrow s + \eta \sum_{i=1}^K w_i c \tanh\left(\frac{R_i s - s}{c}\right), \quad (1)$$

with step size η , weights w_i , and a cap c that bounds how hard any single model can pull in one step. Each model reads the current state through its own frame and writes back a bounded correction; because every model reads a state that every other model has just written to, the models drive each other. This is what makes it a feedback system and not a weighted sum.

The readout is not the settled state alone but the *tail motion* of the trace: whether the update has stopped (consensus), is trembling (competing pulls), or is rotating (ambiguous). Per query q , we also record the *terrain* $\|qJ^\top\|$, where J is the Jacobian of the update map — intuitively, how much the system's stability depends on which question you asked.

The quantity that decides everything is the spectral radius ρ of the coupled update operator. Intuitively ρ is the gain around the loop: $\rho \leq 1$ means pulls damp out and the sheet settles, $\rho > 1$ means they amplify and it does not.

4 Design choices

A system of this kind is determined by three decisions. We list them because they are the axes along which variants differ, and because two of them turned out to be settled by measurement rather than by argument.

1. Wiring — who talks to whom. Whether every model couples to every other, or all models couple to a shared hub. We describe both: the object of Eq. (1) is all-to-all through the shared state; the anchor construct of §6.1 is a star through a hub. [measured] The anchor construct is a star: no model-to-model map exists as an object in the implementation. This has a consequence we did not anticipate and return to in §6.

2. What each model contributes. Two candidates. A model can be represented by the *internal geometry* of its outputs — the pattern of distances among its responses to a probe set, which requires no correspondence between models' coordinates and is unchanged if you rotate or

relabel a model’s output space. Or it can be represented by its *response to perturbation* — how outputs move when inputs are nudged, which carries a direction from input to output but requires both extra access (you must perturb) and a shared labelling of channels across models.

We built the first. The second composes and carries direction, which are real advantages for a router, and we drafted a formalism around it. [retracted] The argument we gave for preferring it was mathematical (§6) and that argument does not survive measurement. What remains is an empirical question — does direction improve routing? — which we have not tested and which we no longer think is a good reason to build.

3. The objective, and what stays outside it. The system minimises one quantity; the diagnostic must be read *outside* that quantity. This is our one architectural commitment and it survives everything below: if the diagnostic enters the objective, the system can improve the diagnostic by optimising it, and the number stops meaning anything. [argued] We previously argued that this separation was forced by a mathematical obstruction rather than chosen; see §6 for the retraction. The rule stands; its justification is now a design discipline rather than a theorem.

5 What the object does

All numbers below are from `explore.py` in the repository at $D = 8$, with the plain average as the null throughout. The stance was exploratory and fixed in advance: report what the object does, do not relabel a no-op as a result, and label injected structure as injected.

Regime	coupled ρ	terrain	vs. average
Convergent frames (agree)	0.955	flat, 0% contested	ties (honest no-op)
Conflicting frames (diverge)	1.090	95% amplifying	terrain differs
Identical control	1.000	flat, 0% contested	ties (correct)
Real modalities (ViT / MiniLM / audio / time-series)	0.962	std 0.032	ties, no-op

Table 1: Behaviour by regime. Real-modality terrain variation (0.032) is *below* the identical-input control (0.051), i.e. no conflict signal is present.

5.1 It is a genuine feedback object, not averaging in disguise

[measured]

We fixed one non-negotiable criterion before building: *if the coupled flow cannot reach $\rho > 1$ on conflicting inputs, it is averaging with extra steps*. Averaging is structurally capped at $\rho \leq 1$ — it cannot amplify, because its output is a convex combination of its inputs. The object reaches $\rho = 1.090$ on conflicting frames and $\rho = 0.955$ on convergent ones. It passes.

We stress this because it is the check that most such systems do not state. A mechanism that looks elaborate but is provably equivalent to a weighted sum is common; naming in advance the measurement that would expose it is cheap and we recommend it.

5.2 It no-ops honestly on real models

[measured]

Four real, independently trained models across four modalities settle to consensus at $\rho = 0.962$. Their terrain is flat — below the identical-input control. On real input the object returns what the average returns.

This is the expected outcome and it is consistent with the representation-convergence literature (§2): independently trained models on related data *do* learn compatible representations, so they do not fight. Our mechanism needs them to fight.

5.3 One behaviour the average lacks — but it is silent where it matters

[measured]

On conflicting frames the terrain is contested: 95% of query directions amplify. On agreeing or identical frames it is flat. The plain average returns the same consensus point either way, so this is a real distinction the average is blind to.

But it is non-vacuous only when frames genuinely conflict, and real modalities do not reach that regime. The signal exists and is silent on real input.

5.4 The payoff does not fire

[measured]

The motivating payoff was to recover competing interpretations from the tremble rather than collapsing them — holding a superposition instead of picking a winner. On a fair test (a hidden binary distinction carried by one frame, neutral query), the object recovers the correct branch at 0.550; the average null is 0.562. Both are near chance and the object does not beat the average. On conflicting frames the tail read is an unresolved pull, not a clean branch-carrying tremble.

This is the finding that determines the object’s current status. The mechanism is faithful and the terrain signal is real, but the thing that would have made it more than averaging on real data does not happen.

5.5 The result that does not need any of this

[measured]

Fusing complementary modalities beats the best single modality: stitched $R^2 = 0.445$ against best-single 0.337. This is the genuinely useful finding in the program and it is independent of the object above — it is coverage, not coupling. We state it here so the paper is not read as claiming the object is responsible for it.

6 The anchor construct, and two pre-registered probes

6.1 The anchor construct

An earlier version of the system couples models not directly but through a shared set of anchor points. Each model is represented by the internal geometry of its output cloud; each model is coupled to the anchors by an entropic semi-relaxed Gromov–Wasserstein plan; the anchors themselves are placed by barycentric averaging and are self-sizing (atoms are added while they clear a noise floor and pruned below it). A diagnostic — the *cycle cost* — measures how far the system is from returning to where it started after traversing a loop through the anchor.

That diagnostic separates structured from scrambled inputs by a factor of 20.4 on the canonical substrate, and we treated it as the system’s validated component. We had also given it a mathematical interpretation: that the cycle cost is a *topological* quantity (a cohomology class), and that this is why the diagnostic must live outside the objective — an obstruction to gluing cannot itself be a term in the thing it obstructs.

We tested that interpretation. Predictions were written down and committed before any code ran (repository commit 9b24e8e).

6.2 Probe A: is the transport structure what the interpretation requires?

[measured]

The topological reading requires that the maps between models be invertible and that they satisfy a consistency condition on three-way overlaps. Measured:

- The couplings are 128×6 transport plans — non-square, rank at most 6. No inverse is formed anywhere in the implementation.
- Round-trip error $\|T_{ji} \circ T_{ij} - I\| = 0.996$; invertibility fails in 0 of 20 pairs. Even a model’s own self-loop is 0.995 from the identity.
- The three-way consistency condition fails in 0 of 10 cases (0.996), and the number of strict three-way overlaps is **zero** — the wiring is a star through the hub, so three-way overlaps do not exist by construction.

[retracted] The topological reading is not licensed. The cycle cost is a *metric* quantity — a distance from closure — not a cohomology class. The measurement survives; the interpretation does not. Consequently the argument in §4 for keeping the diagnostic outside the objective is a design discipline, not a theorem, and the mathematical case for preferring a direction-carrying representation is withdrawn.

6.3 Probe B: is the diagnostic an artefact of the solver?

[measured]

A separation of 20.4 could reflect something the models carry, or merely the order in which the solver updates things. Measured:

- **Cold floor:** the validated path performs no warm-starting, so 20.4 *is* the cold number. Across seeds: 20.4, 12.1, 15.6, 63.3, 42.4 (median 20.4), all far above the pre-registered threshold of 3.
- **Warm-start contribution:** 0.5%. Warm-starting is not manufacturing the signal.
- **Model update order:** invariant to permutation (coefficient of variation 1.95×10^{-16}).

[open] One axis was not measurable. The order in which the solver alternates between updating couplings and updating anchors is forced by data dependency and not exposed as a parameter; testing it would require re-parameterising the solver, which was out of scope for a measurement-only session. We therefore certify “not a solver artefact” only on the axes the existing implementation exposes, and flag the remaining one rather than claiming it.

7 Current status

- The coupled-feedback object is faithful, honestly no-ops on real models, carries one agree-versus-conflict signal that is silent on real input, and its motivating payoff performs at chance. [measured]
- The anchor construct’s diagnostic is robust, cold, reproducible, and model-carried on every measurable axis — but it is a metric quantity, not a topological one, and one solver axis remains untested. [measured], [open]

- The direction-carrying alternative representation has lost its mathematical justification and has not been built. [retracted]
- The useful result in the program (coverage-union) does not depend on any of the above. [measured]

We have not run a systematic prior-art sweep against the 2024–2026 model-merging and equilibrium-routing literature. Until we do, we make no originality claim beyond the specific combination described in §2.

8 Limitations

- **Scale.** The regime experiments run at $D = 8$ with four models. Nothing here speaks to behaviour at realistic scale.
- **One task family.** The real-modality evaluation uses a single set of four models on one dataset. The no-op result may not generalise, though the mechanism by which it arises (representation convergence) is well documented elsewhere.
- **Two systems, one paper.** The coupled-feedback object and the anchor construct are different systems that coexist in the repository; the anchor construct’s history is preserved on a separate branch. We report both because both were measured, but the results should not be read as pertaining to a single artefact.
- **One untested solver axis** (§6).
- **No originality claim.** See above.
- **Negative results are not proofs of impossibility.** That our payoff did not fire does not establish that no such mechanism can.

9 Reproducibility and disclosure

All code, the pre-registration, the probe scripts, and the raw result tables are in the repository. `explore.py` reproduces every number in §5; `PREREG-PROBES.md`, `PROBE-A-RESULTS.md` and `PROBE-B-RESULTS.md` contain the probe predictions and outcomes, with the predictions committed before the runs.

Use of language models. This manuscript was drafted with the assistance of a large language model (Anthropic Claude), which also assisted in writing the probe scripts and in the design discussions that produced the object. All experimental numbers are outputs of code in the repository and were not generated by a language model. The author is responsible for all claims. Where a previously held interpretation is withdrawn, the withdrawal is stated explicitly rather than silently corrected.

References

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